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DATA SCIENCE

Assignment : 1

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Submitted to

Prof. Anjana. J

Anjana

Submitted by

Fathuma Hathiya. A

Roll no: 19

Sg MCA

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P.O.E

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1. Given the following data on a certain set of patients seen by a doctor. Can the doctor conclude that a person having chills, fever, mild headache and sothroat running nose has flu?

Chills	Running Nose	Headache	fever	Has flus
Y	N	mild	Y	N
Y	Y	No	N	Y
Y	N	strong	Y	Y
N	Y	mild	Y	Y
N	N	No	N	N
N	Y	strong	Y	Y
N	Y	strong	N	N
Y	Y	mild	Y	Y
Y	N	mild	Y	?

Consider the above certain set of patients where we can assume set of patients as $P_1, P_2, P_3 \ldots P_8$ we have to find that P_9 patient has flus or not.

In have P_9 patient has some instances.

chills - Y, fever, Y, Headache - mild, Running nose - N.

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So lets calculate,

probability of patient having flu: 5/8
probability of patient not having flu: 3/8.

Now we want to calculate probability based on instance.

chills	Y	N
Y	3/5	1/3
N	2/5	2/3

Running Nose	Y	N
Y	4/5	1/3
N	1/5	2/3

fever	Y	N
Y	4/5	1/3
N	1/5	2/3

Headache	Y	N
mild	2/5	1/3
No	1/5	1/3
Strong	2/5	1/3

for finding P_1 has flu or not.

let's calculate

$P_1 = P(\text{has flu} = Y) \times P(\text{chills} = Y) \times (\text{Running Nose} = N)$

$\times P(\text{Headache} = \text{mild}) \times P(\text{fever} = Y)$

$= 5/8 \times 3/5 \times 1/5 \times 2/5 \times 4/5$

$= 0.024$.

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$P_{\sim B}(N) = P(\text{Has flu} = N) \times P(\text{chills} = Y) \times P(\text{Running nose} = N) \times P(\text{Headache} = \text{mild}) \times P(\text{fever} = Y)$

$= 3/8 \times 1/8 \times 2/3 \times 1/3 \times 1/3$

$= 0.009$

lets find the obtained solution is correct or not

$$P(Y) = \frac{P_{\sim B}(Y)}{P_B(Y) + P_{\sim B}(N)} = \frac{0.021}{0.021 \times 0.009} = 0.727$$

$$P(N) = \frac{P_{\sim B}(N)}{P_{\sim B}(Y) + P_{\sim B}(N)} = \frac{0.009}{0.021 + 0.009} = 0.273$$

$P(Y) + P(N) = 0.727 + 0.273 = 1$

So the obtained solution is correct. we can conclude that.

probability of the patient P_A is recalled as having flu with 0.727 for given instances having chills, fever, mild headache without running nose. Also comparing to probability of not having flu.

Consider a training data set consisting of the of world. Each unit has 3 features names 'sun', "fly", and 'crawl' let the possible values of their

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features be as follows.

Swim - fast, slow, no

fly - long, short, Rarely, no

Crawl - Yes, no

for simplicity each unit is classified as "Animal", "Bird" or "fish". Let the training data set be as the table below. Use naive layer algorithm to classify a particular species if its features are (slow, rarely, no)

Sl.no	Swim	fly	Crawl	class
1	fast	no	no	fish
2	fast	no	yes	Animal
3	slow	no	no	Animal
4	fast	no	no	Animal
5	no	short	no	Bird
6	no	short	no	Bird
7	no	Rarely	no	Animal
8	slow	no	yes	Animal
9	slow	no	no	fish
10	slow	no	yes	fish
11	no	Long	no	Bird
12	fast	no	no	Bird
13	slow	Rarely	no	?

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Consider the training data set of given above each unit has some features they are.

Swim - fast, slow, no
fly - long, short, Rarely, no
Crawl - yes, no

Also here, each unit are classified into three Animal, Bird and fish

We have to find out which classificateon would be 18th under given features like Swim = Slow, fly = Rarely and crawl = no

Probability of class = fish = 3/12
Probability of class - Animal = 5/12
Probability of class - Bird = 4/12

now we have to find probability based on features

crawl	fish	Animal	Bird
yes	1/3	2/5	0/4
no	2/3	3/5	4/4

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fly	fish	Animal	Bird
Long	0/3	0/5	1/4
Short	0/3	0/5	2/4
No	3/3	4/5	1/4
Rarely	0/3	1/5	0/4

Swim	fish	Animal	Bird
fast	1/3	2/5	1/4
slow	2/3	2/5	0/4
No	0/3	1/5	3/4

To find out which classification

Let's find out

probability = P~B (fish) = P (class = fish) × P (Swim = slow)

× P (fly = Rarely) × P (crawl - No)

= 3/12 × 2/3 × 0 × 2/3 = 0//

P~B (Animal) = P (class = Animal) × P (Swim = slow) × P (fly

Rarely) × P (crawl = No)

= 5/12 × 2/5 × 1/5 × 3/5 = 0.02

P~B (Bird) = P (class = Bird) × P (Swim = slow) ×

P (fly - Rarely) × P (Crawl = No)

= 4/12 × 0 × 0 × 4/4 = 0.

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find the obtained solution is correct or not.

$$P(\text{fish}) = \frac{P_{\sim B}(\text{fish})}{P_{\sim B}(\text{fish}) + P_{\sim B}(\text{Animal}) + P_{\sim B}(\text{Bird})} = \frac{0}{0+0\cdot 0\cdot 2+0}$$

$$= \underline{\underline{0}}.$$

$$P(\text{Animal}) = \frac{P_{\sim B}(\text{Animal})}{P_{\sim 0}(\text{fish}) + P_{\sim B}(\text{Animal}) + P_{\sim B}(\text{Bird})}$$

$$= \frac{0\cdot 02}{0+0\cdot 0\cdot 02+0} = \underline{\underline{1}}.$$

$$P(\text{Bird}) = \frac{P_{\sim B}(\text{Bird})}{P_{\sim B}(\text{fish} + P_{\sim B}(\text{Animal}) + P_{\sim B}(\text{Bird})}$$

$$= \frac{0}{0+0\cdot 2+0} = \underline{\underline{0}}$$

$$P(\text{fish}) + P(\text{Animal}) + P(\text{Bird})$$

$$= 0+1+0 = \underline{\underline{1}}$$

So the obtained solution is correct.

probability of 15th fauna is resulted as Animal classification with 1 for given features. ∴ Saim= Shark, fly- Rarely, Crawl= No, while comparing to other classification.

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